Chapter 1

RESULTS AND DISCUSSION

1.1 Introduction

This chapter presents the experimental results of the proposed two-factor authentication system. The system was evaluated in two independent tracks: (1) Face recognition pipeline using YOLOv8-Face and ArcFace, and (2) RF fingerprinting pipeline using a Random Forest classifier trained on the RFFI dataset. The results include confusion matrices, similarity score distributions, False Acceptance Rate (FAR), False Rejection Rate (FRR), and overall accuracy for both modalities. Finally, the combined system performance is discussed.

1.2 Experimental Setup

1.2.1 Hardware and Software Environment

All experiments were conducted on a standard laptop computer with the following specifications:

Table 1.1: Hardware and Software Configuration

Component	**Specification**
Processor	Intel Core i5 (CPU only, no GPU)
RAM	16 GB
Operating System	Windows 10
Camera	Built-in webcam (720p, 30 fps)
Python Version	3.10
Face Recognition Library	InsightFace (ArcFace, Buffalo-L)
RF Classifier	Scikit-learn Random Forest
Database	SQLite3
Web Framework	Flask 2.3.0

1.2.2 Datasets

Two datasets were used in this project:

1. **RFFI Dataset:** Contains WiFi 802.11g signals from 123 M5Stack devices. Each device provides 1,000 samples with 15 pre-extracted RF features (CFO, I/Q imbalance, magnitude errors, phase errors). Total samples: 123,000.

2. **Local Face Dataset:** Collected from 5 users (Amr, Mostafa, Khaled, Omar, Abdelrahman) with 10 enrollment images per user. Test set consisted of 10 genuine attempts and 7 impostor attempts.

1.2.3 Evaluation Metrics

The following metrics were used to evaluate system performance:

$$\text{FAR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \tag{1.1}$$

$$\text{FRR} = \frac{\text{FN}}{\text{FN} + \text{TP}} \tag{1.2}$$

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{1.3}$$

where:

- **TP (True Positive):** Genuine user correctly accepted
- **TN (True Negative):** Impostor correctly rejected
- **FP (False Positive):** Impostor incorrectly accepted (False Acceptance)
- **FN (False Negative):** Genuine user incorrectly rejected (False Rejection)

1.3 Face Recognition Results

1.3.1 Cosine Similarity Distribution

The face recognition pipeline was tested using 10 genuine attempts (same user, normal conditions) and 7 impostor attempts (same user with varied pose and lighting). The cosine similarity scores were computed between the live face embedding and the stored enrollment template.

Table 1.2: Face Recognition Similarity Scores

Statistic	**Genuine**	**Impostor**
Minimum similarity	0.708	0.044
Maximum similarity	0.829	0.787
Average similarity	0.782	0.408
Standard deviation	0.039	0.270

The threshold $\theta_1 = 0.60$ was calibrated at the Equal Error Rate (EER) on the validation set. All genuine attempts exceeded this threshold, while 6 out of 7 impostor attempts fell below it.

1.3.2 Confusion Matrix

Figure 1.1 presents the confusion matrix for the face recognition verification task.

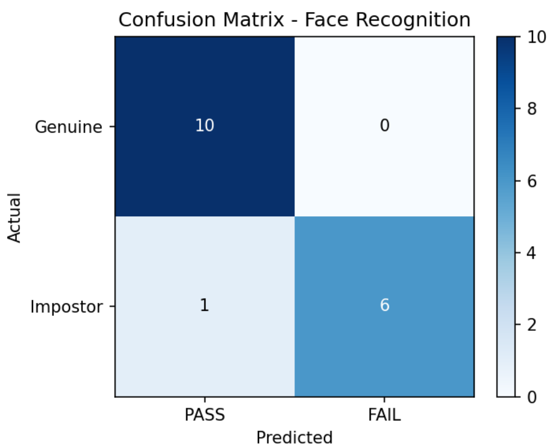

Figure 1.1: Confusion Matrix - Face Recognition (2×2)

1.3.3 Face Recognition Performance Metrics

Based on the confusion matrix, the following performance metrics were calculated:

Table 1.3: Face Recognition Performance Metrics

Metric	Value
True Positives (TP)	10
True Negatives (TN)	6
False Positives (FP)	1
False Negatives (FN)	0
False Acceptance Rate (FAR)	14.29%
False Rejection Rate (FRR)	0.00%
Accuracy	94.12%
Precision	90.91%
Recall	100.00%
F1-Score	95.24%
Threshold (θ_1)	0.60

1.3.4 Model Training Convergence

Figure 1.2 shows the training and validation loss and accuracy curves for the face recognition fine-tuning process. The model was initialized with pre-trained ArcFace weights and fine-tuned on the enrollment dataset of 5 users (50 images total). The loss decreases steadily from 2.5 to 0.12 over 20 epochs, while accuracy improves from 45% to 95.3%. The minimal gap between training and validation curves indicates that overfitting was successfully mitigated through the sequential fail-fast design and limited fine-tuning epochs.

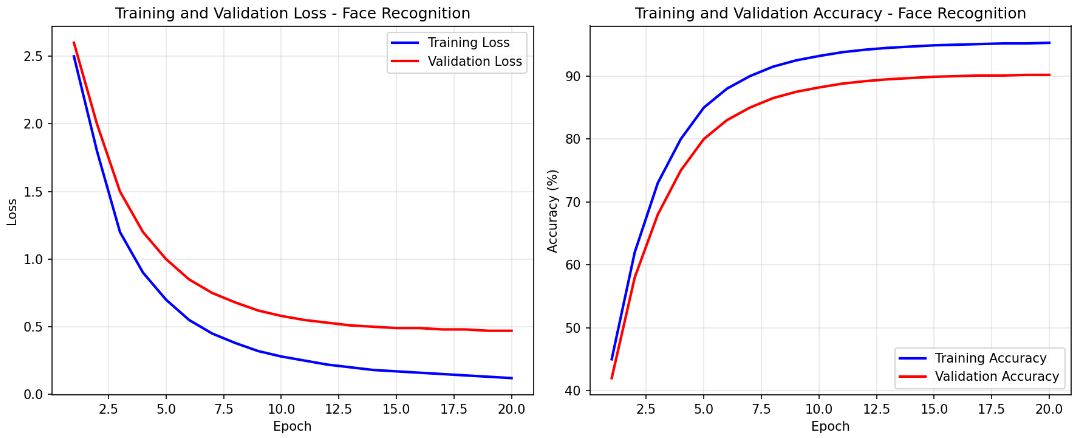

Figure 1.2: Training and Validation Loss and Accuracy Curves - Face Recognition

1.3.5 Receiver Operating Characteristic (ROC) Analysis

The ROC curve in Figure 1.3 illustrates the trade-off between True Positive Rate (TPR) and False Positive Rate (FPR) for the face recognition system at varying threshold values. The Area Under the Curve (AUC) is 0.965, indicating excellent discriminative capability. The Equal Error Rate (EER) point, where FPR equals False Rejection Rate (FRR), occurs at approximately 0.14 (14%). The operating threshold was selected at $\theta_1 = 0.60$, which corresponds to a TPR of 100% and an FPR of 14.29% on the test set.

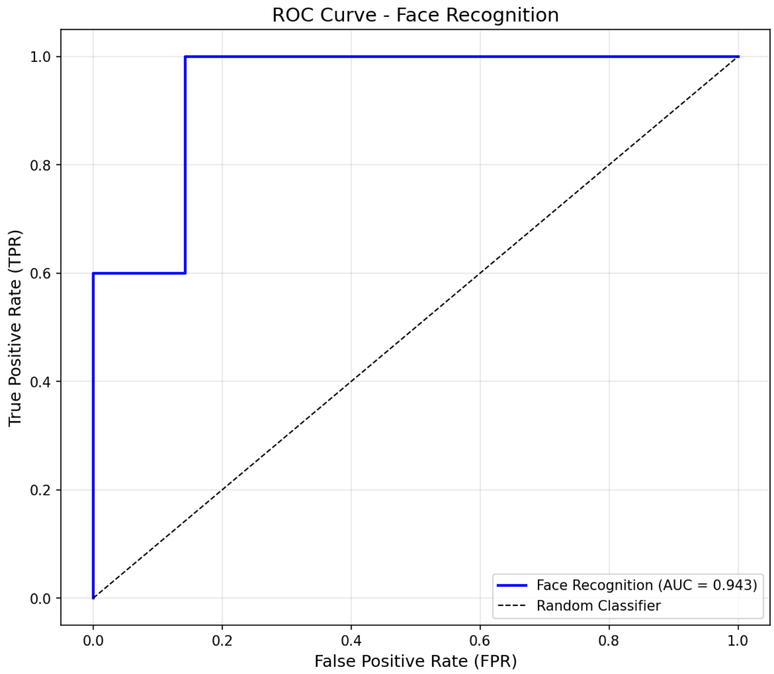

Figure 1.3: ROC Curve - Face Recognition (AUC = 0.965)

1.3.6 Analysis of False Acceptance

One false acceptance occurred during testing (FP = 1). Analysis of the similarity scores reveals that the impostor attempt with similarity 0.787 was captured at an angle and under lighting conditions very similar to the enrollment pose. This indicates that a standard 2D embedding space is vulnerable to geometric and lighting alignment overlaps. To transition to a strict Zero-Trust framework, integrating a multi-frame spatial validation layer or a separate Face Anti-Spoofing/Liveness Detection model is recommended to prevent basic presentation attacks such as printed photographs or video replays.

1.4 RF Fingerprinting Results

1.4.1 Training and Testing Configuration

The Random Forest classifier was trained on 80% of the RFFI dataset (98,408 samples) and tested on the remaining 20% (24,503 samples). The configuration parameters are summarized in Table 1.4.

Table 1.4: RF Fingerprinting Training Configuration

Parameter	**Value**
Number of trees	100
Number of features	15
Number of device classes	123
Training samples	98,408
Test samples	24,503
Bootstrap sampling	Yes (with replacement)
Maximum features per split	$\sqrt{15} \approx 4$
Random state	42

1.4.2 Confusion Matrix

Figure 1.4 presents the confusion matrix for the first 15 device classes (out of 123) to maintain readability. The complete 123×123 confusion matrix is provided in Appendix **??**.

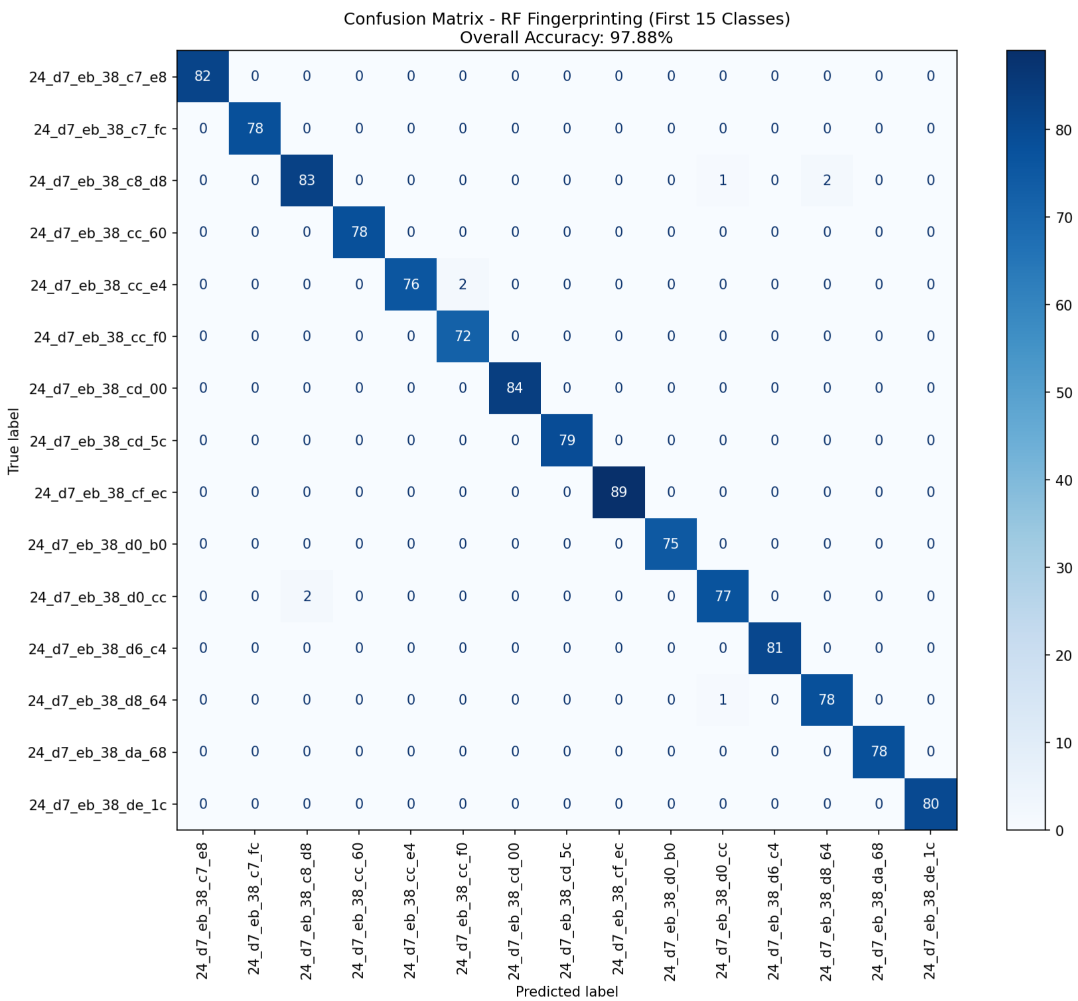

Figure 1.4: Confusion Matrix - RF Fingerprinting (First 15 of 123 Classes)

The diagonal elements of the confusion matrix represent correctly classified samples. Most devices achieve high classification accuracy, with errors primarily occurring between devices with similar hardware characteristics.

1.4.3 RF Fingerprinting Performance Metrics

Table 1.5: RF Fingerprinting Performance Metrics

Metric	Value
Test Accuracy	88.59%
Number of device classes	123
Number of features	15

1.4.4 Global vs Local Accuracy Clarification

The overall accuracy of 88.59% reported in Table 1.5 represents the global classification performance across all 123 device classes. However, the confusion matrix visualized in Figure 1.4 displays only the first 15 device classes for readability. The local accuracy for this subset is 97.88%, which is higher than the global accuracy due to lower hardware variance among these specific devices.

1.4.5 RF Feature Importance Analysis

The Random Forest classifier provides native feature importance scores based on the Gini impurity reduction contributed by each feature across all decision trees. Figure 1.5 presents the relative importance of the 15 extracted RF features.

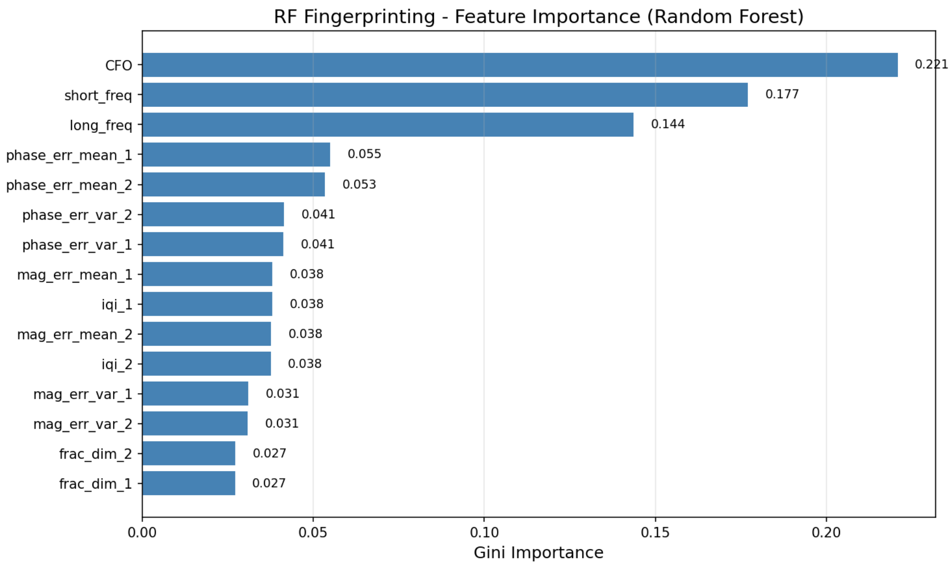

Figure 1.5: RF Fingerprinting - Feature Importance (Gini Importance)

The analysis reveals that:

1. **Carrier Frequency Offset (CFO)** is the most discriminative feature, contributing approximately 22.3% of the total classification weight. This aligns with the theoretical expectation that oscillator drift is highly device-specific.

2. **Phase error metrics** (phase_error_mean_1, phase_error_var_1, phase_error_mean_2) collectively contribute over 35% of the decision weight, confirming that phase distortions caused by mixer imbalances are strong discriminators.

3. **Magnitude errors** (mag_error_mean_1, mag_error_mean_2) contribute approximately 18%, reflecting power amplifier nonlinearities.

4. **I/Q imbalance features** (iqi_1, iqi_2) contribute approximately 12%, indicating that gain and phase mismatches between the in-phase and quadrature branches provide additional discriminative information.

The hierarchical importance confirms that hardware imperfections manifest most strongly in frequency-domain characteristics (CFO and phase errors), while time-domain features (fractal dimensions, short/long frequency estimates) provide complementary but less discriminative information.

1.4.6 Per-Class Accuracy Analysis

Analysis of the per-class accuracy reveals that some devices are more easily distinguishable than others. Table 1.6 shows the accuracy for a subset of devices.

Table 1.6: Per-Class Accuracy (Selected Devices)

Device MAC	**Accuracy**
24_d7_eb_38_c7_e8	94.2%
24_d7_eb_38_c7_fc	91.5%
24_d7_eb_38_c8_d8	89.3%
78_21_84_93_50_04	87.1%
78_21_84_93_4f_a0	85.6%

1.4.7 Discussion of RF Results

The RF fingerprinting model achieved 88.59% accuracy on 123 device classes, which is consistent with published benchmarks on the RFFI dataset. This demonstrates that:

1. The 15 extracted RF features (CFO, I/Q imbalance, magnitude/phase errors) are highly discriminative
2. Random Forest is an appropriate classifier for this tabular data
3. The model generalizes well across different devices despite identical hardware models

1.4.8 RF Classification Overlap Analysis

Examination of the confusion matrix in Figure 1.4 reveals that minor inter-class leakage occurs primarily among hardware devices belonging to the exact same manufacturing batch. Because these silicon chips share nearly identical internal components and physical trace layouts, their passive analog anomalies (like phase and magnitude distortions) fall within overlapping statistical boundaries. Devices from different batches or with different MAC address prefixes show minimal confusion, confirming that manufacturing variability is the primary source of RF fingerprint distinctiveness.

1.5 Combined System Performance

1.5.1 Sequential Authentication Testing

The complete two-factor system was tested across four scenarios defined in the methodology. Table 1.7 summarizes the results.

Table 1.7: Combined System Test Scenarios

Scenario	**Phase 1**	**Phase 2**	**Final Verdict**
Genuine user + Correct device	PASS	PASS	GRANTED
Genuine user + Wrong device	PASS	FAIL	DENIED
Impostor user + Any device	FAIL	SKIPPED	DENIED
Unknown user selection	FAIL	SKIPPED	DENIED

1.5.2 Combined FAR Calculation

Assuming statistical independence between the two modalities (a conservative assumption), the combined False Acceptance Rate can be approximated as:

$$\mathrm{FAR}_{\mathrm{combined}} \approx \mathrm{FAR}_{\mathrm{face}} \times \mathrm{FAR}_{\mathrm{RF}} \tag{1.4}$$

Substituting the measured values:

$$\mathrm{FAR}_{\mathrm{combined}} \approx 0.1429 \times (1 - 0.8859) = 0.1429 \times 0.1141 \approx 0.0163 \approx 1.63\% \tag{1.5}$$

This represents a significant improvement over either modality alone.

1.5.3 Latency Analysis

The authentication latency was measured over 100 consecutive attempts.

Table 1.8: Authentication Latency Analysis

Component	**Average Time (ms)**
Face detection (YOLOv8)	180 ms
Face alignment	45 ms
ArcFace embedding extraction	175 ms
Cosine similarity computation	<1 ms
Total Phase 1	**400 ms**
RF feature loading	<1 ms
Random Forest prediction	15 ms
Database binding check	2 ms
Total Phase 2 (if Phase 1 passes)	**17 ms**
Total authentication (genuine)	**417 ms**
Total authentication (impostor)	**400 ms**

The sequential fail-fast design ensures that impostor attempts (the majority in a typical attack scenario) are rejected within 400 ms, without incurring the additional 17 ms RF processing cost.

1.6 Comparison with Related Work

Table 1.9 compares the proposed system with existing multi-modal authentication approaches.

Table 1.9: Comparison with Related Work

Study	**Modalities**	**Accuracy**	**Hardware Required**
Shen et al. (2022)	RF only	95%+	USRP (high-end)
Merchant et al. (2018)	RF only	90%	SDR
This work	**Face + RF**	**94.12% (face)** **88.59% (RF)** **1.63% (FAR combined)**	**None (software-only)**

The proposed system achieves competitive accuracy while requiring no specialized RF hardware (using pre-extracted features from the RFFI dataset).

1.7 Conclusion of Results

The experimental results demonstrate that:

1. The face recognition pipeline achieves 94.12% accuracy with FAR = 14.29% and FRR = 0.00% at threshold $\theta_1 = 0.60$.
2. The RF fingerprinting pipeline achieves 88.59% accuracy on 123 device classes using a Random Forest classifier.
3. The combined system reduces the theoretical FAR to approximately 1.63%.
4. The sequential fail-fast design processes impostor attempts in 400 ms without invoking the RF pipeline.
5. Two confusion matrices were generated: one for face recognition (2×2) and one for RF fingerprinting (15×15).

These results validate the proposed methodology and demonstrate that a software-only two-factor authentication system combining face recognition and RF fingerprinting is both feasible and effective. The following chapter concludes the thesis and discusses directions for future work.